PawMatch Development Test Plan

This is a pre-development plan. Each test states what will be checked, the data that will be used, the expected result and the analysis evidence that justifies the test. The Pass column is left blank until the test is completed during development.

Source codes

E1 Dogs Trust competitor research E2 RSPCA competitor research E3 Petfinder competitor research
Q number Client interview question S number Stakeholder survey finding

Authentication

Test	Description and justification	Test data or input	Expected result	Source	Pass
A1	Check that the registration page loads with all required fields and buttons.	Open Register.	Name, email, password and confirmation fields are visible.	Q6 S6	
A2	Create an account with valid data to prove registration works.	Name: Test Adopter; valid email; age 24; valid password.	One user record is created and the user can continue to the home page.	Q9 Q12	
A3	Check invalid and duplicate registration data so bad accounts are not stored.	Blank field; invalid email; duplicate email; age 17; mismatched passwords.	A clear error is shown and no new user record is created.	Q9 Q12 Q15	
A4	Check password security so plain-text passwords are never stored.	Register two users with the same valid password.	Both hashes differ and neither stored value equals the password.	Q12	
A5	Check login with correct and incorrect details.	Valid email/password, then wrong password and unknown email.	Valid login opens Home; invalid attempts show a generic error and remain on Login.	Q12	
A6	Check role-based access so adopters cannot use administrator pages.	Log in as adopter and enter an admin URL; repeat as admin.	Adopter is redirected; administrator is allowed access.	Q5 Q12	

Navigation and session handling

Test	Description and justification	Test data or input	Expected result	Source	Pass
N1	Check protected pages cannot be opened without logging in.	Clear the session and open Home, Questionnaire, Browse and Account URLs.	Every protected page returns to Login without revealing private data.	Q12	
N2	Check the main navigation and session are consistent.	Log in and open each menu item in turn.	Each control opens the correct page and keeps the same signed-in user.	E1 E2 E3 S6	
N3	Check logout clears the session.	Log out, use Back, then enter a protected URL.	The user stays logged out and protected pages return to Login.	Q12	

Administrator animal records

Test	Description and justification	Test data or input	Expected result	Source	Pass
D1	Add a valid animal so the database has a complete record for matching.	Nova; dog; Labrador; age 2.5; female; medium; activity 4; care fields; description.	One animal record is stored and all values are shown correctly.	Q2 Q5 Q15	
D2	Check boundary and invalid animal data so incomplete records are rejected.	Blank name; name lengths 1, 50 and 51; age -1; activity outside 1-5.	Valid boundaries are accepted; invalid values show an error and are not stored.	Q2 Q15	
D3	Edit an animal and check the saved record remains current.	Change Nova's activity and description, then save and refresh.	Only the selected record changes and the new values remain after refresh.	Q5 Q15	
D4	Delete an animal safely so accidental deletion is prevented.	Choose Delete, cancel once, then confirm.	Cancel changes nothing; confirmation removes the animal and matching results no longer include it.	Q5 Q12 Q15	

Animal browsing

Test	Description and justification	Test data or input	Expected result	Source	Pass
B1	Check an empty animal list is handled clearly.	Use a database with no available animals.	The page shows a helpful empty-state message and does not crash.	E1 E2 E3	
B2	Check search and filters help adopters narrow the list.	Search a full/partial name; filter by species and breed; use an unknown value.	Only matching animals appear; no-match input gives a clear empty result.	E1 E2 E3 Q8	
B3	Check an animal profile shows the correct current record.	Open two different animals from the results.	Each profile shows the selected animal's correct name, traits, care needs and description.	E1 E2 E3 Q2	

Questionnaire and matching algorithm

Test	Description and justification	Test data or input	Expected result	Source	Pass
Q1	Check required questionnaire answers so incomplete data is not used.	Leave one required answer blank and submit; then complete it.	Incomplete answers are rejected; the complete questionnaire is accepted.	**Q1** **S3** **S5**	
Q2	Check questionnaire boundaries and invalid values.	Try minimum, maximum, just-outside and wrong-type values for hours, space and cost.	Valid boundaries are accepted; invalid values show an error and are not saved.	**Q1** **Q10** **S5**	
Q3	Check questionnaire answers are stored against the correct user.	Create two users with different answers, save, log out and return.	Each user sees only their own saved answers.	**Q12** **Q15**	
M1	Check the main compatibility score with a hand-calculated case.	Use one controlled adopter profile and one animal with known traits.	The program score equals the manual calculation and stays between 0 and 100.	**Q10** **Q15** **S5** **S7**	
M2	Check a poor match produces a low score and an explanation.	Use deliberately conflicting housing, time, experience and activity answers.	The score is low and the explanation identifies the important mismatches.	**Q3** **Q10** **S5** **S9**	
M3	Check several animals are ranked correctly.	Run the same questionnaire against at least three controlled animal profiles.	Animals are ordered from highest to lowest score with no duplicate or missing result.	**Q10** **Q14** **S7**	
M4	Check unavailable animals are not recommended.	Mark the highest-scoring animal unavailable and run matching again.	The unavailable animal is excluded and the remaining results are re-ranked.	**Q5** **Q15** **S9**	

Whole-system checks

Test	Description and justification	Test data or input	Expected result	Source	Pass
I1	Check the complete adopter journey after the modules are connected.	Register, log in, complete questionnaire, browse, view a profile and open results.	Every step succeeds and the same user and data are kept throughout.	**E1** **E2** **E3** **Q14** **S6** **S7**	
I2	Check the complete administrator journey.	Log in as admin, add, edit, find and delete a disposable animal.	All database changes are correct and are reflected on adopter-facing pages.	**Q5** **Q12** **Q15**	
I3	Check invalid actions do not crash the program or corrupt data.	Repeat key forms with blank, wrong-type, overlong and unsupported values.	The program stays usable, explains the problem and writes no invalid data.	**Q12** **Q15**	
I4	Check basic usability on a narrow screen.	Open the main journeys at a mobile-width viewport using keyboard and pointer input.	Text remains readable, controls remain usable and navigation does not cover content.	**Q6** **Q13** **S6**	

Analysis source key

The short codes used in every test row point back to evidence already collected in the analysis. They show why each test is relevant to the intended user and stakeholder needs.

Competitor research

Code	Evidence source	How it informed the plan
E1	Dogs Trust	Clear browsing, animal profiles and adoption information.
E2	RSPCA	Species filters, welfare information and clear navigation.
E3	Petfinder	Search, filters, detailed profiles and ranked recommendations.

Client interview

Code	Evidence used
Q1	Important lifestyle factors including space, time, cost and experience.
Q2	Information that should be stored and displayed for each animal.
Q3	Warnings and explanations when a match is unsuitable.
Q5	Administrator add, edit and delete functions with current records.
Q6	The system should be easy to use.
Q8	Search and filtering are needed.
Q9	Registration information and validation.
Q10	How compatibility factors should be weighted.
Q12	Security, user roles and protected access.
Q13	The interface should work on the stakeholder's available devices.
Q14	What the user should see after compatibility results are calculated.
Q15	Accurate scoring and current, detailed animal profiles are essential.

Stakeholder survey

Code	Evidence used
S3	Lifestyle compatibility was the most common priority.
S5	Living space, time, exercise, cost and experience were important factors.
S6	Ease of use and clear navigation were important.
S7	Compatibility scores would influence decisions.
S9	Animal welfare must remain more important than a high score.